PriceShield — Real-World Consumer Protection Co-Pilot

An on-device AI system for point-of-sale retail overcharging, chilling-fee de-escalation, and packaging compliance.

1. Executive Summary

PriceShield is an intelligent, point-of-sale consumer rights co-pilot built to protect everyday Indian shoppers against retail overcharging (MRP violations, illegal chilling fees, and promotional bundle splitting). Unlike bureaucratic post-purchase grievance portals that require 15-minute multi-step forms, PriceShield empowers consumers **in real time (< 3 seconds)** directly at the checkout counter with 0ms barcode intelligence, on-device computer vision, bilingual statutory audio de-escalation, and native evidence preparation.

> ***Core Innovation:*** *PriceShield does not merely record grievances. It equips the consumer with verifiable statutory legal backing (Rule 18(2) Legal Metrology Rules) to de-escalate and resolve unfair charges on the spot peacefully.*

2. The Problem & Market Landscape

Across millions of Kirana stores, highway dhabas, bus stands, and cinema halls in India, three consumer abuses persist daily:

1. Transit & Chilling Surcharges: Retailers charge ■5–■25 over MRP for cold drinks, water bottles, and milk. Retailers cite refrigeration costs, which is explicitly prohibited under Rule 18(2) of Legal Metrology (Packaged Commodities) Rules 2011.

2. Multipack & Promo Splitting: Retailers tear open promotional packs marked *'Not for individual sale'* or *'Buy 3 Get 1 Free'* and sell individual units at standalone MRP, illegally pocketing consumer discounts.

3. The High-Friction Redressal Gap: 95% of consumers stay silent because confronting vendors causes ugly public arguments, while filing on government portals takes 15–20 minutes of tedious paperwork for a ■10–■20 dispute.

3. Strategic Differentiation: PriceShield vs. Government NCH (1915)

A common strategic query is: *'Since the government already provides National Consumer Helpline (NCH 1915), why would consumers use PriceShield?'*
PriceShield is not a duplicate or competitor to NCH; it is the **Consumer-Side Intelligence and Evidence Preparation Layer**.

Dimension	Government NCH (1915)	PriceShield Platform
Point of Interaction	Post-purchase at home on a web portal	**At the shop counter in real-time (< 3 sec)**
Filing Friction	15–20 mins, 20+ form inputs, manual bill uploads	**10-second flow: 1-tap scan, auto GPS, camera OCR**
At-the-Counter Resolution	Zero instant support at the shop	**Bilingual Audio Law Card de-escalates peacefully**

Packaging Compliance	None; cannot inspect multipacks or promo offers	**On-Device Vision AI flags *'Not for individual sale'***
Evidence Reliability	Often discarded due to unverified user text	**High-res proof, geostamp, Cloudinary zoom**
Escalation Output	Internal departmental queue only	**WhatsApp merchant notice + Formal NCH docket draft**

4. Technical Justification: Why Is PriceShield 'AI-Powered'?

PriceShield deliberately avoids using 'AI' as marketing buzz. It implements four targeted machine learning and NLP components designed to function reliably on mobile hardware without expensive API dependencies:

A. Computer Vision Neural Network (Google ML Kit):
Rather than simple 1D barcode decoding, PriceShield runs on-device deep learning vision models. This extracts text labels from curved bottles, reflective metallized foils (chips pouches), and wrinkled labels under uneven lighting conditions.

B. Intelligent Entity & Compliance Extraction:
Physical packaging contains extreme noise: *'BATCH 104 PKD 11/24 FSSAI LIC 1001202 USP 0.28/g MRP Rs. 20.00'*. PriceShield applies heuristic pattern classifiers to filter date/batch noise, isolate true statutory MRP, and flag violations like *'Not for loose sale'* and *'■5 off'*.

C. Dynamic Neural Audio Synthesis (Bilingual TTS):
To de-escalate confrontations, PriceShield dynamically synthesizes natural bilingual legal speech (Hindi & English). It binds live dispute numbers (shop price vs. statutory MRP) with Rule 18(2) legal phrasing to present an authoritative voice directly from the phone.

D. Automated Legal Notice Synthesis (NLP Drafting):
Transforms unstructured incident facts (overcharge delta, store location, product name, evidence links) into formal legal notices and NCH-ready dispute dockets citing the Consumer Protection Act 2019.

5. The Zero-Cost (■0) Self-Growing Data Engine

Acquiring millions of product barcodes without paying expensive GS1 / DataKart enterprise API licensing is solved via a **4-Stream Free Flywheel** (modeled after Vivino & Waze):

Stream 1 — Bundled Offline Catalog: 175+ high-velocity Indian FMCG products (Amul, Parle, Maggi, Sting, Tata Salt) bundled directly in the client app for **instantaneous 0ms offline lookup**.

Stream 2 — Open Food Facts API (ODbL License): 30,000+ Indian FMCG products queried via real-time backend fallback with a strict 2.5s timeout and automatic database caching.

Stream 3 — On-Device ML Kit Auto-Harvest: When an unlisted barcode is scanned, Google ML Kit reads the printed label from camera footage and permanently seeds the central database.

Stream 4 — 'Shield Scout' Gamification: Users earn **+50 ShieldPoints** for verifying new products in their neighborhood, creating a crowd-sourced, self-expanding catalog.

Storage Economics: *Each product record requires ~180 bytes in PostgreSQL. Supabase's free tier (500 MB) accommodates over* ***25,00,000 (2.5 Million) FMCG products at ■0 infrastructure cost****.*

6. Production Architecture & Tech Stack

Layer	Technology	Key Role & Responsibility
Mobile App	Flutter 3.x (Dart)	Cross-platform client, ML Kit Vision OCR, In-Memory 0ms Cache
Backend API	Node.js + Express (Render)	RESTful micro-modules, token verifier, Open Food Facts fallback
Database	Prisma ORM + PostgreSQL (Supabase)	Indexed barcode tables, complaint evidence relation, audit logs
Media Cloud	Cloudinary	Unsigned evidence uploads with interactive zoom/pan photo viewer
Auth	Firebase Auth + Custom JWT	Verified credentials with support for unauthenticated guest scanning

7. Current Milestones & Live Verification Status

- **Live Database:** 175 real FMCG daily-use products seeded and live in Supabase PostgreSQL.
- **Speed Engine:** Instant 0ms in-memory cache and 2.5s strict timeout fallback verified.
- **Test Coverage:** 34/34 Flutter integration/unit tests passed; 16/16 backend Vitest test suites passed.
- **UI Polish:** Interactive proof viewer with pinch-to-zoom, consumer-friendly toasts, and zero developer jargon.
- **Next Roadmap:** Packaging Compliance Scanner (Multipack / Promo Fraud) & Merchant 1-Click Settlement Links.